Lecture 13

- Diagonalization
- Symmetric Matrices
  - Eigen properties
  - Spectral Thm
  - Optimization interpretation of eigenvalues

Diagonalization (why eigenbases are useful)

Any square matrix $A \in \mathbb{M}_{n\times n}$ induces a linear transformation of $\mathbb{R}^n$, $x \mapsto Ax$. (1)

$L : \mathbb{R}^n \to \mathbb{R}^n$ such that $\left.\begin{array}{l} L(v_1+v_2) = L(v_1)+L(v_2) \\ L(\alpha\cdot v) = \alpha\cdot L(v). \end{array}\right\}$

Any linear transf. can be written in form (1).

Indeed, let $e_1 \dots e_n$ be the stand. basis and $v \in \mathbb{R}^n$.

Then $L(v) = L(\alpha_1 e_1 + \dots + \alpha_n e_n) = \alpha_1 \underbrace{L(e_1)}_{\in \mathbb{R}^n} + \dots + \alpha_n \underbrace{L(e_n)}_{\in \mathbb{R}^n}$

Let $L(e_j) = a_{1j} e_1 + \dots + a_{nj} e_n = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} = a_j$

$$\Rightarrow L(v) = \alpha_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix} + \dots + \alpha_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{nn} \end{pmatrix} = \underbrace{[a_1 \dots a_n]}_{A} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \underbrace{A\cdot\alpha}. \quad (2)$$

matrix representation of L in $\{e_i\}$

coordinates of v in $\{e_i\}$

coordinates of $L(v)$ in $\{v_i\}$

Matrix A in (2) is called a matrix representation of L.

Of course, this representation depends on the chosen basis of $\mathbb{R}^n$.

Q: How does the matrix representation of L depend on basis?

Let $\{e_1 \dots e_n\}$ be the stand. basis and $\{v_1 \dots v_n\}$ be another basis of $\mathbb{R}^n$.

Let $u \in \mathbb{R}^n \Rightarrow \begin{cases} u = \alpha_1 e_1 + \dots + \alpha_n e_n = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \alpha & \text{coord. of } v_1 \text{ in } \{e_i\} \\ u = \beta_1 v_1 + \dots + \beta_n v_n = \beta_1 \begin{pmatrix} v_1 \end{pmatrix} + \dots + \beta_n \begin{pmatrix} v_n \end{pmatrix} = \underbrace{[v_1, \dots, v_n]}_{V} \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}}_{\beta} \end{cases}$

$\Leftarrow \boxed{\alpha = V\beta}$ (*)

coord of u in $\{e_i\}$

coord of $\{v_j\}$ in $\{e_i\}$

coord of u in $\{v_i\}$

- In $\{e_i\}$: $L(u) = A\cdot\alpha = \alpha'$ ← mat. rep. of L in $\{e_i\}$
- In $\{v_i\}$: $L(u) = \tilde{A}\beta = \beta'$ ← mat rep of L in $\{v_i\}$

$$(*) \Rightarrow \underbrace{A\alpha}_{\alpha'} = V \underbrace{\tilde{A}\beta}_{\beta'} \Rightarrow AV\beta = V\tilde{A}\beta \ \forall\beta \Rightarrow AV = V\tilde{A} \text{ or } \boxed{\tilde{A} = V^{-1}AV}$$

This motivates the following definition:

Def Two matrices $A, B \in \mathbb{M}_{n\times n}$ are called similar if $\exists V$ nonsing : $B = V^{-1}AV$.

Remark: Similar matrices represent the same linear transf.

Def A matrix $A \in \mathbb{M}_{n\times n}$ is diagonalizable if it is similar to a diagonal matrix:

(3) $\quad \Lambda = V^{-1}AV \quad$ or $\quad A = V\Lambda V^{-1}, \quad \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

Q: If A is diagonalizable, what is the meaning of V, Λ?

(3) $\Leftrightarrow \quad AV = V\Lambda \quad \Leftrightarrow \quad A[v_1 \dots v_n] = [v_1 \dots v_n]\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

k^{th} column: $\quad Av_k = \lambda_k v_k$.

Thus, $\{\lambda_1, \dots, \lambda_n\} = \operatorname{spec} A$ and $v_k \in V_{\lambda_k}$.

Thm A is diagonalizable $\Leftrightarrow$ A is complite.

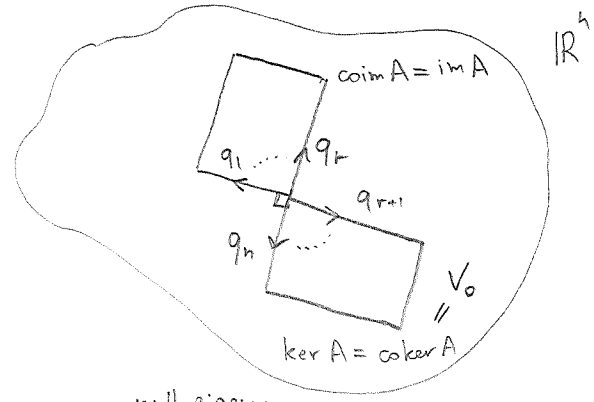
Symmetric Matrices

Let A be a symmetric matrix, $A^T = A$.

- All eigenvalues of A are real, $\operatorname{spec} A \subset \mathbb{R}$
- Eigenvectors associated with distinct eigenvalues are $\perp$
  $v_\lambda \in V_\lambda$ and $v_\mu \in V_\mu$ $(\lambda \neq \mu) \Rightarrow v_\lambda \perp v_\mu$. $(V_\lambda \perp V_\mu)$.
- A admits orthonormal eigenbasis of $\mathbb{R}^n$
  (A is complite, A is diagonalizable).
- A is (semi-) positive definite $\Leftrightarrow \lambda_i > 0 \quad (\lambda_i \geq 0)$

Let A be symmetric.

Fund. Thm:



Let $\underbrace{q_1 \dots q_r}_{\text{assoc. with } \lambda \neq 0}\ \overbrace{\underbrace{q_{r+1} \dots q_n}_{\text{assoc. with } \lambda = 0}}^{\text{null eigenvectors}}$ be an orthonormal eigenbasis of A

$\ker A = \{x : Ax = 0\} = V_0 = \operatorname{span}(q_{r+1}, \dots, q_n)$

- $\{q_{r+1}, \dots, q_n\}$ form an orthon. basis of $\ker A$
- $\{q_1 \dots q_r\}$ form an orthon. basis of $\operatorname{im} A$

Let $Q = [q_1 \dots q_n]$ be an orthogonal matrix.

$\Rightarrow AQ = Q\Lambda \Rightarrow A = Q\Lambda Q^T$

Spectral Thm If A is symmetric, then $\exists$ orthogonal Q and diagonal Λ:

$$A = Q\Lambda Q^T$$

$\Lambda = \operatorname{diag}(\lambda_1 \dots \lambda_n), \quad \lambda_i \in \operatorname{spec} A$

$Q = [q_1 \dots q_n], \quad Aq_i = \lambda_i q_i, \quad q_i \in V_{\lambda_i}$

Optimization interpretation of spec A

Let A be symmetric,

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be its eigenvalues.

<u>Goal</u> to represent $\lambda_i = \max_x f(x)$ for some f.

Intuition comes from a diagonal matrix.

Let $\Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, $\underbrace{q_\Lambda(x)}_{\text{quadratic form}} = x^T \Lambda x = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2$.

Maximizing q_Λ over $\mathbb{R}^n$ does not make much sense:

$$\max_{x \in \mathbb{R}^n} q_\Lambda(x) = \begin{cases} +\infty & \lambda_1 > 0 \\ 0 & \lambda_1 \leq 0 \end{cases}$$

But let's max. it over $S^n = \{x : \|x\|^2 = \sum_{i=1}^n x_i^2 = 1\}$

- $q_\Lambda(x) = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2 \leq \lambda_1 (x_1^2 + \dots + x_n^2) = \lambda_1$
- $q_\Lambda(e_1) = \lambda_1 \quad (e_1 \in V_{\lambda_1})$

$$\Rightarrow \max_{x \in S^n} q_\Lambda(x) = \lambda_1$$

Similarly, $\min_{x \in S^n} q_\Lambda(x) = \lambda_n$.

Let A be a general symmetric matrix.

$\xRightarrow[\text{Spec.}]{\text{Thm}} A = Q \Lambda Q^T$.

$$q_A(x) = x^T A x = \underbrace{x^T Q}_{y^T} \Lambda \underbrace{Q^T x}_{y} = y^T \Lambda y$$

So, we have a map: $\underset{S^n}{\underset{\cap}{x}} \mapsto y = Q^T x$.

$y \in S^n$

$\|y\|^2 = \langle y, y \rangle = y^T y = x^T \underbrace{Q Q^T}_{I} x = x^T x = 1$

Moreover, since Q is nonsingular, $x \leftrightarrow y$ is a 1-to-1 map.

Therefore,

$$\max_{x \in S^n} q_A(x) = \max_{y \in S^n} y^T \Lambda y = \lambda_1.$$

<u>Thm</u> If A is a symmetric matrix, then

$$\boxed{\lambda_1 = \max_{x \in S^n} q_A(x)} \quad \text{and} \quad \boxed{\lambda_n = \min_{x \in S^n} q_A(x)}$$

maximum is achieved on $v \in S^n \cap V_{\lambda_1}$

minimum is achieved on $v \in S^n \cap V_{\lambda_n}$.

<u>Q</u>: What about $\lambda_2, \dots, \lambda_{n-1}$?

Intuition again comes from the diagonal case:

- $\max_{x \in S^n} q_\Lambda(x) = \lambda_2$

  $x = (0, x_2, \dots, x_n) \Leftrightarrow x \in \operatorname{span}(e_1)^\perp$

- $\max_{x \in S^n} q_\Lambda(x) = \lambda_i$

  $x = (0 \dots 0, x_i, \dots, x_n) \Leftrightarrow x \in \operatorname{span}(e_1 \dots e_{i-1})^\perp$

In general:

$\lambda_1 \leq \dots \leq \lambda_n$

$v_1 \dots v_n$ corresponding orthonormal basis.

$$\boxed{\lambda_i = \max \left\{ q_A(x) \,/\, \|x\| = 1,\ x \in \operatorname{span}(v_1, \dots, v_{i-1})^\perp \right\}}$$